

Web Application Development - Context and Memo

Duration: 15 minutes.

* Indicates required question

1. Full Name *

2. Student ID *

3. Which of the following is true about the Context API? *

2 points

Check all that apply.

- ☐ Context should be used when you need to share global data.
- ☐ Context requires you to pass props down manually at every level of the component tree.
- ☐ Context is the only way to pass data through the component tree.
- ☐ Context was introduced by React as a way to resolve the props drilling problem.

4. Assuming that the default theme for the application is "light", what should be the default value passed to the createContext call? * 2 points

Check all that apply.

- ☐ undefined
- ☐ null
- ☐ The string "light"

5. One of the parts of the context injected into the application is a function called toggleTheme. Assuming that the theme is held in some local state as a string that can be either "light" or "dark". What should be the exact implementation of the toggleTheme function? * 1 point

Mark only one oval.

- ☐ toggleTheme: () => setTheme(theme === "light" ? "dark" : "light")
- ☐ toggleTheme: (theme) => setTheme(!theme)
- ☐ toggleTheme: () => setTheme(theme === "light" ? "light" : "dark")

6. What of the below scenarios are valid for choosing context instead of local state? * 1 point

Check all that apply.

- ☐ The locale or language that should be used in the application's text.
- ☐ The current selection of a group of radio buttons.
- ☐ The visibility state of an alert that overlays into the whole application.

7. What is the problem of props drilling? *

1 point

Check all that apply.

- ☐ Components having to pass down props all the way to the children that need to consume them.
- ☐ Components not knowing the local state of their parents.
- ☐ Components receiving more props than they should.

8. When creating a new piece of application state, what is the bare minimum of React APIs you would need to define it? * 1 point

Mark only one oval.

- ☐ Context and local state.
- ☐ Context and props.
- ☐ Context, props and local state.

9. What happens when the value prop of the Context Provider changes? *

1 point

Mark only one oval.

- ☐ All the consumer components re-render with the updated value.
- ☐ The whole component tree under the Context Provider gets re-rendered.
- ☐ The Context Provider component gets recreated.

10. What happens when you wrap a component with the React.memo API, such as `React.memo(Component)`. * 1 point

Check all that apply.

- ☐ React provides a performance optimization.
- ☐ The component never gets updated no matter if there was a change in its local state or the props it receives.
- ☐ Whether the component should re-render could be determined by some custom logic that uses the previous props and the current props.

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